

Will AI Replace Doctors?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

2.1

AI EXPOSURE · SURGERY

where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



Before you read this

The choice is already made — you're in a medical degree or training, or out of one and into the job hunt and the Match. So this guide skips the "should you study medicine" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

What this is. A facts document, not a pep talk. It sets out what's known about where medicine sits in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

The good news, up front. Already in it is not too late — not close. Medicine isn't one destination, it's a wide spread of them, from 2.1 to 4.9, and almost none of your exposure is fixed yet. Where you land inside that range is decided mostly by the specialty you choose and the skills you build — moves still in front of you. The second half of this guide is those moves.

And one thing specific to medicine. For the first time, the specialty you choose carries an AI dimension it didn't a few years ago. The profession as a whole is the best-protected in this index, but the specialties are no longer moving together — and that single choice is now where most of the decision lives. Steering it deliberately is exactly what this guide is for.

The short answer

Medicine remains one of the safest paths available, and patient-facing practice holds the strongest protection stack in this entire index. But for the first time, choice of specialty carries an AI dimension. Whatever stage you're at, a specialty choice is ahead of you or newly behind you — and the specialties are no longer moving together.

Surgery scores 2.1 — the lowest score of any career we measure. Radiology scores 4.9 and is the fastest-moving protected career we track. Same degree, same training, a 2.8-point gap.

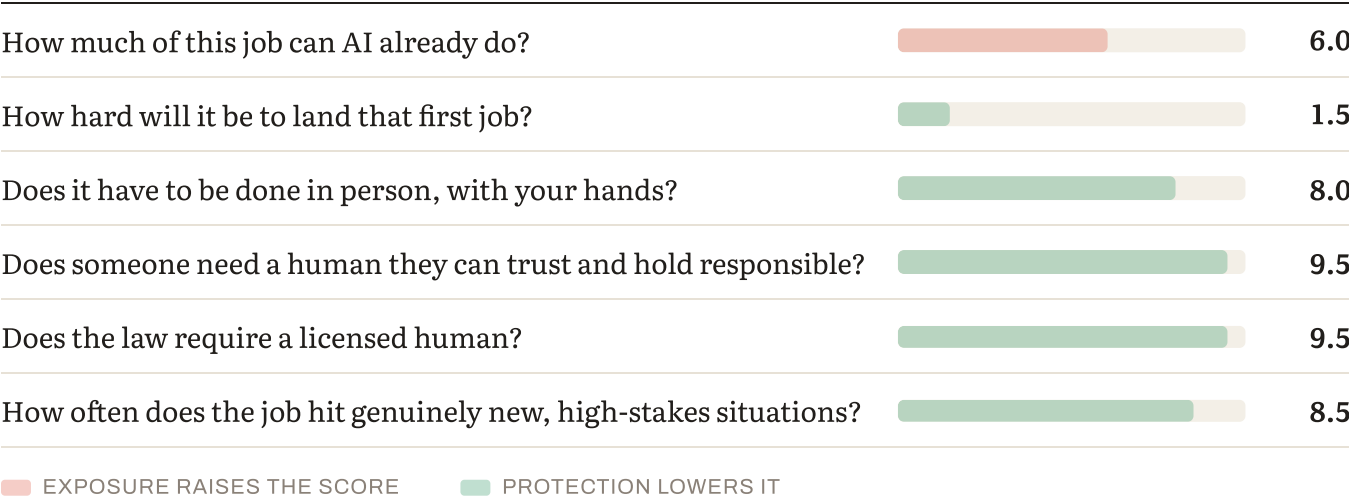
The scores

SPECIALTY	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Surgery	1.8	1.9	2.1	+0.3	Very low
Emergency medicine	2.0	2.2	2.5	+0.5	Very low
Primary care / family medicine	2.4	2.6	2.9	+0.5	Low
Anesthesiology	2.7	2.9	3.1	+0.4	Low
Psychiatry	3.1	3.4	3.7	+0.6	Low
Pathology	3.8	4.3	4.7	+0.9	Moderate
Radiology	3.9	4.4	4.9	+1.0	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, bedside nursing scores 2.8, and entry-level software development scores 8.1.

Read the table this way: every specialty in medicine sits below the index median, so the profession as a whole is exceptionally well protected. But the spread has widened from 2.1 points in 2023 to 2.8 today, and it widens every year we measure it. Medicine is not becoming riskier. It is becoming less uniform — and where you land inside that spread is decided mostly by the specialty you steer toward.

Why medicine scores low — the six factors



We score every profession in this index against the same six questions, with the same weights. Here's how primary care answers them — the reference point for patient-facing medicine.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

A substantial share — and it is the layer around the medicine, not the medicine.

Clinical documentation, letters, coding, record summarization, guideline and drug-interaction lookup, and increasingly triage and risk stratification. Ambient systems that draft the consultation note are already in routine use, and most physicians regard them as a relief rather than a threat.

What is not automating is the physical examination, the decision made under genuine uncertainty, and the conversation with a family when the news is bad.

This is the one factor where specialties genuinely diverge, and the radiology question further on covers that in detail.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

The most protected on-ramp we measure anywhere in this index.

Across almost every other profession, AI is absorbing the junior tasks first — the grunt work that entry-level roles were built from, and that people historically used to build judgment. The professions survive; the paths into them collapse.

Medicine is structurally immune. Residency is mandated, funded, structured and legally required — a doctor cannot exist without supervised years. Whatever else changes about the profession, its apprenticeship is written into the system in a way no employer can quietly erode.

The way through, here, isn't about surviving the on-ramp — medicine's is the one entry route in this index that holds. It's about what you steer toward once you're on it. The divergence is entirely in specialty choice: the patient-facing and procedural specialties (surgery 2.1, emergency medicine 2.5, primary care 2.9) hold their protection, while the diagnostic, screen-based ones (pathology 4.7, radiology 4.9) carry more exposure. That's a set of concrete choices, not a hope — and the repositioning section and the career-hunting checklist later in this guide lay them out.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

For most specialties, substantially yes.

Examining a patient. Operating. Delivering a baby. Resuscitating someone. These are irreducibly physical, and robotics is nowhere near them.

This is also where the internal spread begins: a surgeon rates near the top of this factor and a radiologist near the bottom, and that single difference does more work than any other in separating a 2.1 from a 4.9.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Absolutely, and in a form that is unusually hard to delegate.

Nobody wants an algorithm to tell them they have cancer. And when a clinical decision goes wrong, a licensed human carries the responsibility — legally, professionally and morally. That accountability attaches to a person, not to a system, and no advance in capability transfers it. Which means the fastest way to become protected is to get closer to the decision you own.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Comprehensively, and more thoroughly than almost any other profession.

Licensure, registration, prescribing authority, scope-of-practice rules and malpractice liability all mandate a qualified human. This protection moves at the speed of legislation and public trust, not at the speed of technology — which is why we rank it the most durable protection in the entire framework.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Constantly.

Patients present atypically as a rule. They arrive with several conditions interacting, deteriorate unpredictably, and don't match the textbook. This is precisely where AI remains weakest — genuine novelty, where the pattern doesn't fit and being wrong is expensive.

What the trend shows

We retrospectively scored medicine for 2023 and 2025 using this same methodology. The interesting finding is not the movement — it's the divergence.

Patient-facing medicine: 2.4 → 2.6 → 2.9. Up 0.5 points in three years, and nearly all of it administrative automation. The physical, trust, licensing and judgment scores have not moved at all.

Diagnostic specialties: 3.9 → 4.4 → 4.9. Up a full point — the fastest-moving protected career we score anywhere.

For comparison over the same period, an entry-level software developer moved 1.8 points. Medicine is drifting; software is sprinting. But within medicine, radiology is moving at twice the rate of the specialties around it.

The gap between the safest and most exposed specialty has grown from 2.1 points to 2.8. Three years ago you could reasonably say "medicine is safe" and stop there. That is no longer a complete answer.

How durable is the protection?

Every "AI can't do X" boundary has proven temporary. It couldn't do nuance, then it could. Couldn't do images, then it could. So the useful question isn't whether medicine's protections hold today — they clearly do — but how long each is likely to last.

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Regulatory / licensing	Strongest available	Most durable	Moves at the speed of law. No capability advance repeals a licensing requirement or malpractice liability.
Human trust / accountability	Very strong	Durable	Socially rooted. People want a human present at frightening moments, and someone must answer for the outcome.
Judgment under novelty	Strong	Eroding	Diagnostic support and risk stratification are advancing — currently assisting, not deciding.
Physical / embodied	Strong for most specialties	Most fragile long-term	The protection physical AI directly targets. Surgical robotics is assistive today; the horizon is longer than a career.

The honest read: you'll practice for roughly forty years. Over that span, the two protections that will still be standing are the ones rooted in law and society rather than in what the technology can do. Physical protection is the strongest shield today and the one most likely to weaken within a career.

Note also which protections are specialty-dependent. Licensing and accountability protect every doctor equally. Physical presence and novelty do not — and that is precisely why the specialties are pulling apart, and why the direction you steer matters more than where you happen to start.

How the role will actually change

Not "will medicine exist" — it will. The useful question is what the job looks like in ten years.

WAS THE JOB	IS BECOMING THE JOB
Writing the note after the consultation	Ambient systems draft it; the physician verifies and corrects
Recalling the guideline	Decision support surfaces it; the physician judges whether it fits this patient
Reading the scan	Supervising the AI read, adjudicating the ambiguous cases, owning the decision
Documentation consuming the evening	More time with patients — the part most doctors came for
Triage by experience	Risk stratification flags the deteriorating patient earlier

The pattern across this index is that AI absorbs the routine, repeatable layer first. In creative and knowledge work that layer was often the craft itself. In medicine it is the paperwork, which is the leading cause of physician burnout. Automating it returns time to patients rather than removing the practice.

Two honest caveats. Verification of AI-generated notes and reads becomes a real clinical responsibility — a wrong record is a patient-safety issue, not an administrative one. And efficiency gains can be captured as higher patient loads rather than returned as time, which is a funding decision rather than a technological one.

Human strengths that will matter most

Ranked by how well each resists the coming decade, specific to medicine:

1. **Physical examination and clinical intuition** — noticing the patient who looks wrong before the numbers say so. The hardest thing to replicate and the most valuable thing a doctor builds.
2. **Judgment under genuine uncertainty** — the atypical presentation, the interacting conditions, the case with thin precedent and high cost of error.
3. **Accountability** — being the human who owns the decision. Durable, legally rooted, and the reason a qualified person stays in the loop.
4. **Communication at the hardest moments** — breaking bad news, negotiating a

treatment plan, being trusted by a frightened family. 5. **Verification and discernment** — knowing when an AI-generated note, read or recommendation is subtly wrong. A genuinely new core skill, and one barely taught anywhere yet.

Notably absent: speed of recall and breadth of memorized knowledge. These were the traditional markers of a strong medical student and are the fastest-depreciating. If your current value is mostly on that second list, the work is to shift it onto the first.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The ambient documentation, decision support and diagnostic tools will turn over many times across a forty-year career. The doctors who compound are the ones who stay genuinely curious about the medicine and re-tool fastest, rather than defending the one way of working they trained on. In a profession where the automatable layer is the paperwork rather than the practice, that flexibility is itself a form of protection — and increasingly it's what the informatics and leadership paths are looking for: the trajectory, not the current tool.

Other things to consider about this job

Everything above measures one thing: how exposed medicine is to AI. That's useful, and it's deliberately narrow. It says nothing about whether the work is rewarding, whether the training is survivable, or who this suits.

All of that matters more to a real decision than the score does. So here it is, in four parts.

What's genuinely good about it

WHAT DOCTORS THEMSELVES SAY

The satisfaction picture in medicine is genuinely two-sided, and both sides are real.

Physician job satisfaction sits at 77% and has risen every year since 2022, according to the AMA's national survey. **Satisfaction is highest among physicians in their first five years out of training and those twenty years or more in — the difficult stretch is the middle.**

What they report as rewarding is remarkably consistent:

- **Being trusted at the most consequential moments of someone's life** — the relationship, not the transaction

- **Intellectual depth** — medicine remains one of very few careers where the learning genuinely never stops
- **Mastery of something difficult** — competence in a demanding craft, earned over years
- **Breadth of choice** — few qualifications open onto so many genuinely different working lives
- **Autonomy at senior level**, and in many systems the option of independent practice

THE PRACTICAL CASE

- **Exceptional job security.** Physician shortages are projected across most developed health systems, and the training pipeline is constrained by funding rather than demand.
- **A portable qualification**, though re-licensing across countries is more onerous than for nursing.
- **Strong earnings**, with wide variation by specialty and system.
- **Genuine specialty choice** — someone who wants procedures, or continuity, or acute pressure, or research, can find a version of this career that fits.

What's genuinely hard about it

In medicine, automation is not the biggest risk to this career. The length and cost of getting there, and burning out once you have, are the real ones.

THE LENGTH

This is the single most under-appreciated fact about medicine, and it is rarely held concretely in mind.

From starting a medical degree to independent practice is typically **ten to fifteen years** depending on specialty and country. That period includes sustained examination pressure, repeated relocation for training posts in most systems, and years of working under supervision at a stage when peers in other careers are already established.

THE COST

In systems where medical education is paid for privately, the debt is substantial and it compounds during years of relatively low training-grade income. This is a genuine constraint on who can realistically enter the profession, and it should be discussed openly rather than assumed away.

BURNOUT AND MORAL INJURY

Physician burnout is high — but it has been **falling for three straight years**, and this is one place the popular narrative is out of date. The AMA's 2025 survey of nearly 19,000 physicians across 106 health systems found 41.9% reporting at least one burnout symptom, down from 43.2% in 2024 and 48.2% in 2023.

It remains meaningfully above the general working population, and it is very unevenly spread: emergency medicine, urological surgery and hematology/oncology all sit near or above 49%. Around 31% of physicians report intending to leave their organization within two years.

The most-cited drivers are administrative burden, loss of autonomy, and the specific distress of being unable to provide the care you believe a patient needs because of time, staffing or system constraints.

That last one has a name — **moral injury** — and it is distinct from being tired. It is what makes people leave a profession they still believe in.

EARLY-CAREER HOURS

Training-grade hours remain demanding in most systems, with night and weekend work through the years when many people are also forming families.

Who this work suits — and who it doesn't

Fit predicts staying far better than aptitude does. Medicine tends to suit people who:

- **Can hold uncertainty without freezing.** Much of clinical practice is deciding well with incomplete information. Some people find that energizing; others find it corrosive.
- **Get calmer when things go wrong.** A genuine and identifiable trait, hard to teach, and disproportionately valuable.
- **Are genuinely interested in people, not just in biology.** The science is the entry ticket; the practice is relational.
- **Can sustain a long horizon.** The rewards arrive late. Someone who needs visible progress every year will find the training years hard.
- **Tolerate being wrong in public.** Medicine involves error, review, and being corrected by colleagues, throughout a career.

Equally honest about who finds it hard:

- People who need **control over their own schedule** will struggle for a decade or more.

- People who need **certainty and closure** find the ambiguity difficult.
- People drawn primarily by **status or parental expectation** are heavily over-represented among those who leave.

Squeamishness, for what it's worth, is almost never the barrier people expect — nearly everyone acclimatizes faster than they think.

What else is moving this market

The AI score measures task exposure. It cannot see supply, demand, funding or the politics of health systems — and those often matter more.

Training capacity is the constraint, not demand. Medical school places and residency positions are capped by funding rather than by applicant quality or patient need. That protects the profession's scarcity, and it also means competition for entry stays fierce regardless of how many doctors a country needs.

Satisfaction varies more between specialties than the profession-level numbers suggest. Psychiatry reports 83% job satisfaction and obstetrics 81.2%, while hospital-based specialties — emergency medicine, radiology, anesthesiology — sit lowest at 74.8% and perform below benchmark on most wellbeing measures. That is a reminder that the setting matters as much as the specialty, and neither is visible in an AI score.

Shortages are structural in most systems, particularly in primary care, psychiatry and several procedural specialties. This is a profession where the labor market has been tight for decades and is projected to stay tight.

Working conditions are the volatile part. Pay, hours, administrative load and autonomy move with health-system funding and politics rather than with technology. Understand that the technology outlook for medicine is stable and the conditions outlook is not.

And one specialty-specific note. Radiology deserves flagging here because its labor market and our score point in opposite directions — see the radiology question, below.

The radiology question — where our own score is contested

We flag this because it is the strongest evidence against any number in this index, and it concerns the specialty most people ask about.

The case for our 4.9. Image interpretation is pattern recognition on structured visual data — the single thing AI does best. Around three-quarters of all FDA-authorized AI-enabled medical devices are in radiology, because that is where the technology works. The score has moved a full point in three years, faster than any other protected career we track.

The case against it. The labor market has done the opposite of what the narrative predicts. Roughly half of US radiologist job openings went unfilled in recent years. Imaging volume is growing 3–4% annually against a residency pipeline capped by federal training funding — a supply constraint with nothing to do with AI. Diagnostic radiology filled at about 97% in the 2025 Match, and interventional at essentially 100%. And there is little evidence so far that AI has delivered substantial productivity gains to the specialty.

Radiology has been the poster child for imminent AI displacement for roughly a decade, and the shortage has deepened throughout.

Where we land. We hold the score, and we publish the disagreement alongside it. Exposure to automation is not the same as exposure to unemployment, and in radiology the gap between those two is wider than anywhere else we measure. Bending a number to match an observed outcome would make every trend in this index meaningless.

One subtlety worth carrying. Radiology applications have declined from their 2023 peak, and reporting attributes part of that to student uncertainty about AI. So the fear is already shaping behavior even where the labor market isn't — which is arguably a reason a document like this should exist.

What we're watching: whether radiologist vacancy rates start falling, and whether AI-assisted reporting produces measurable throughput gains. Either would move the score, and we'll say so.

The AI-native advantage — what it actually looks like

The situation, factually: AI is already in clinical settings. As of early 2025 the FDA had authorized 882 AI-enabled medical devices, and the large majority of US hospitals use some form of AI-enabled clinical decision support. Meanwhile most practicing physicians are encountering these tools without any formal training in them.

That gap is the entire opportunity. The technology is deployed. The people who know how to use it well are not — and being already in it, close to the work, is a real advantage here. You can build this fluency inside a placement, a training post or a first job, which is exactly where it counts.

WHAT AN AI-NATIVE DOCTOR ACTUALLY LOOKS LIKE

Not someone who has memorized tools. Someone who can do five things:

- 1. Verify.** Knowing when a generated note, read or recommendation doesn't match the patient in front of them. **This is a clinical skill, not a technical one** — it requires knowing the patient well enough to catch a plausible-sounding error.
- 2. Escalate.** Knowing what to do when the output is wrong: who to tell, how to document it, how to override safely. Institutions are still writing these policies.
- 3. Explain.** Being able to tell a patient, in plain language, what a system contributed and why a human remains accountable. Trust is medicine's deepest protection; someone who can articulate it under scrutiny defends it.
- 4. Recognize the limits.** Understanding at a working level why models produce confident errors, and how algorithmic bias shows up differently across patient populations. This is the ethics dimension and it is increasingly examined.
- 5. Contribute.** Being the person who can say whether a system works in practice. Physicians who can speak to implementation get pulled onto committees, into informatics, and eventually into leadership.

All five are available to you now — not after some future qualification — and on a **dimension where seniority confers no advantage**. Turning them into concrete moves is the rest of this guide.

Where a medical degree can take you later

One of the strongest arguments for medicine is that it doesn't strand anyone. If the clinical path stops suiting you, the qualification remains valuable.

- **Specialty practice** — the main path, with enormous variation in rhythm, pressure and lifestyle
- **Academic medicine and research** — clinical trials, translational research, teaching
- **Clinical informatics** — the bridge between practice and health IT. **The people who understand both the ward and the software are exactly who is needed to implement and supervise the systems described above.**
- **Public health and policy** — population health, epidemiology, government and NGO work

- **Medical leadership** — clinical directorship, medical directorship, health-system management
- **Industry** — pharmaceutical medicine, medical devices, regulatory affairs
- **Medico-legal, occupational health, and expert practice**

The caveat that runs through this whole index applies here too: the further a role sits from the patient, the more its exposure climbs. Informatics and industry roles are genuinely good careers — and they score in the moderate band rather than the low one. That's a reason to move knowingly, not a reason not to — and it's the compass for every choice below.

Repositioning from here

The program-selection stage is behind you. The useful questions now are different — and the good news bears repeating: **almost none of your exposure is fixed yet.** Medicine is a 2.8-point spread from 2.1 to 4.9, and where you land inside it is decided mostly by the specialty you choose and the skills you build — things still in front of you. Nearly all of them point the same way: toward the patient and the decision you own.

If you're still in the degree or training

Three levers, in order of leverage.

1. Specialty choice is the whole game. The divergence is entirely here. The patient-facing and procedural specialties — surgery (2.1), emergency medicine (2.5), primary care (2.9), anesthesiology (3.1), psychiatry (3.7) — hold their protection; the diagnostic, screen-based ones — pathology (4.7), radiology (4.9) — carry more exposure. Every one sits below the index median, so this is choosing between good and better, not safe and unsafe. If you haven't chosen yet, this is the single highest-leverage decision left in the training. If you have, hold it against the scores and the durability table and ask honestly whether it puts a human in the room or a pattern on a screen. And the rule of thumb: **if two specialties genuinely appeal equally, the more physical and more patient-facing one carries more durable protection.**

2. The durable clinical skills, while you're building them anyway. Physical examination and clinical intuition, judgment under genuine uncertainty, and communication at the hardest moments are the least automatable things you'll ever build — and the fastest-depreciating are speed of recall and memorized breadth. Aim your effort at the first list, not the second; it's far cheaper to build in training than to bolt on later, and it's what separates the protected work from the automatable layer around it.

3. Verification, the genuinely new skill. On placement, treat clinical informatics as a real subject, notice which systems each department uses and where seniors work around them, and practise the verification habit: when a tool suggests something, ask what would make it wrong before accepting it. It's a clinical skill, not a technical one, and barely taught anywhere yet — which is exactly why building it now sets you apart.

If you've graduated and you're job-hunting — or choosing in the Match

Here the specialty you match into matters more than it looks, because that single choice is where medicine's whole 2.8-point spread lives.

- **The diagnostic, screen-based specialties are the exposed end.** Radiology (4.9) and pathology (4.7) are the fastest-moving protected careers we track — image reading is pattern recognition on structured data, the single thing AI does best. Choosing one isn't fatal — every specialty is below the index median — but choose it with the AI dimension in view, not by default.
- **Aim at the patient-facing and procedural specialties for the most durable protection.** Surgery (2.1), emergency medicine (2.5), primary care (2.9): irreducibly physical, high-trust, legally accountable work where the protection concentrates.
- **Don't read a plentiful post as a settled answer.** Roughly half of US radiologist openings went unfilled in recent years and diagnostic radiology filled at about 97% in the 2025 Match — the posts are there. Exposure to automation is not the same as exposure to unemployment, and in radiology that gap is wider than anywhere else we measure. Just don't confuse an open door with a low score.
- **Move away from the patient knowingly, not by drift.** Clinical informatics, industry and medico-legal work are genuinely good careers — and they score in the moderate band rather than the low one. The further a role sits from the patient, the more its exposure climbs.

The AI-native move, from where you stand

You can't change the specialty divergence, but you can be the doctor who directs these tools rather than works around them. The five capabilities above — verify, escalate, explain, recognize the limits, contribute — are all available right now, inside a placement, a training post or a first job. The most powerful for someone already in it is the first: be the person who catches the generated note, read or recommendation that's subtly wrong. That's a clinical skill, not a technical one; it requires knowing the patient well enough to catch a plausible error, and it's barely taught anywhere yet — which means seniority confers no advantage on it. You can start building it this rotation, not after some future qualification.

The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the protection in medicine is built here, in these choices and habits, rather than handed to you. The first list is for anyone still in the degree or training; the second is for the Match and job hunt itself.

During the degree / training — build what can't be automated

- **Choose a specialty with durable protection, and choose it knowingly.** The patient-facing and procedural specialties hold their protection best — surgery (2.1), emergency medicine (2.5), primary care (2.9), anesthesiology (3.1), psychiatry (3.7) — while the diagnostic, screen-based specialties carry more exposure — pathology (4.7), radiology (4.9). Every one sits below the index median, so this is choosing between good and better, not safe and unsafe. If two specialties appeal equally, the more physical and more patient-facing one carries the more durable protection.
- **Build the durable clinical skills deliberately.** Physical examination and clinical intuition, judgment under genuine uncertainty, and communication at the hardest moments are the least automatable things a doctor builds — and the ones that separate the protected work from the automatable layer around it. Speed of recall and memorized breadth are the fastest-depreciating; don't mistake them for the skill.
- **Treat clinical informatics as a real subject.** On placement, notice which systems each department actually uses and where senior clinicians work around them — the workarounds show you where a system fails.
- **Practise verification as a habit.** When a tool suggests something — an ambient note, a decision-support prompt, an AI read — ask what would make it wrong before accepting it. This is a clinical skill, not a technical one, and it's barely taught anywhere yet.
- **Feed curiosity and adaptability on purpose.** The ambient documentation, decision support and diagnostic tools will turn over many times across a forty-year career; the doctors who compound are the ones who stay curious and re-tool fastest.
- **Keep the frame wide.** Understand the length before committing — ten to fifteen years to independent practice — and know that adjacent paths (nursing, physiotherapy, paramedicine, dentistry, physician associate, biomedical research) overlap with medicine but differ enormously in training length, cost and daily rhythm.

During the Match and job hunt — aim, don't just take what's open

- **Aim at the durable specialties on purpose.** Weight the AI dimension in the specialty decision, but don't let it dominate — the spread runs 2.1 to 4.9 and every point of it is

below the index median. Where the human is in the room, the protection concentrates; where the work is reading patterns off a screen, it thins.

- **Read the specialty, not just the opening.** The protections rooted in physical presence, trust and accountability attach to the doctor at the bedside, in the theater, in the resus room. A post closer to the patient and to the decision you own is closer to the most durable protection medicine has.
- **Don't grab an exposed-lane role just because it's on offer.** A role being available doesn't make it safe. Roughly half of US radiologist openings went unfilled in recent years and diagnostic radiology filled at about 97% in the 2025 Match — the posts are plentiful, and the specialty still carries the highest AI exposure in medicine at 4.9. Exposure to automation is not the same as exposure to unemployment, so this isn't a reason to avoid it — only a reason not to read an open door as a settled answer on the automation dimension.
- **Move away from the patient knowingly, not by drift.** Clinical informatics, industry and medico-legal work are genuinely good careers — and they score in the moderate band rather than the low one. The further a role sits from the patient, the more its exposure climbs.
- **Ask where AI is heading in the role.** On placement or interview, whether a department is building the job around ambient documentation, decision support and real verification — or waiting to be forced to — tells you if the seat grows or is quietly being hollowed out.
- **Weigh the trajectory over the immediate.** A training post that builds the durable clinical, procedural and verification skills beats one that leaves you doing the automatable layer, however convenient it looks now.

Go deeper — further reading

The evidence:

- **How AI Will Shape the Future of Health Care in 2026** — SullivanCotter. The clearest statement of the pattern this guide describes: AI produces first drafts while clinicians concentrate on validation, interpretation and decision-making. Also covers why radiology has shown earlier proof points than other specialties.
- **Radiology in 2026: The Workforce Crisis Meets the AI Revolution** — Becker's. The two forces acting on diagnostic specialties simultaneously — a genuine shortage against reimbursement compression and accelerating AI adoption. Important because it shows the pressures are not purely technological.

- **AI solutions to the radiology workforce shortage** — *npj Health Systems (Nature)*. Peer-reviewed framing of AI in radiology as a response to a supply shortage rather than a cause of displacement.
- **Physician burnout rate continues to decline, falling to nearly 42%** — *American Medical Association*. The primary source for the wellbeing figures above: nearly 19,000 physicians across 38 states and 106 health systems. Worth reading because the direction of travel contradicts most popular coverage — burnout has fallen three years running while satisfaction has risen.
- **Physician burnout declines overall, but gaps persist across specialties** — *Medical Economics*. The specialty-level breakdown, and the finding that hospital-based specialties underperform on most wellbeing measures.

The counterargument — read this one:

- **Match Day 2026: Radiology programs offer more positions than ever** — *Radiology Business*.

This is the strongest evidence against any score in our index, and it targets our 4.9 directly. Roughly half of US radiologist openings went unfilled in recent years. Imaging volume grows 3–4% annually against a residency pipeline capped by federal funding. Diagnostic radiology filled at about 97% in the 2025 Match; interventional at essentially 100%.

Where we land: we hold that diagnostic work is more exposed than patient-facing work — image reading is what AI does best, and the concentration of FDA authorizations shows where the technology is. But exposure to automation is not exposure to unemployment, and in radiology the gap is wider than anywhere else we score. We publish the number with the disagreement stated rather than adjusting it to fit, and we will report it plainly if vacancy rates start to fall.

Bottom line

Medicine holds the strongest protection stack in this index — comprehensive licensure, deep trust, physical presence in most specialties, legal accountability, and constant novel judgment. Its entry route is the most protected we measure anywhere — which is exactly why being already in it is not a problem. The decisive moves were always going to happen from here.

The risk question is largely settled. The real questions are the ones this career has always posed: ten to fifteen years to independence, the cost, the hours, and whether the vocation is

genuine.

What this index adds is one thing: specialty choice now carries an AI dimension it didn't carry five years ago. The protection concentrates where the human is in the room, and thins where the work is reading patterns off a screen. That divergence is widening every year we measure it.

So the useful advice isn't a verdict on the degree — it's a direction. Steer toward the patient-facing and procedural specialties where the protection concentrates, weighting the AI dimension without letting it dominate. Build the durable clinical, procedural and verification skills deliberately, while the training makes them cheap to acquire. Read the specialty, not just the opening. And build real AI fluency — verification above all — inside whatever role you're in now, because that's the differentiator this profession rewards on a dimension where seniority confers no advantage.

And it's worth restating what the risk analysis can't measure. Doctors report among the highest sense of meaning of any profession measured. Being trusted at the most consequential moments of people's lives, mastering something genuinely difficult, and a career where the learning never stops — very few paths offer that combination at all. The hard parts above are real and shouldn't be minimized. So is that.

Questions worth sitting with

Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.

Where you actually stand

1. Which specialty are you on track for, or aiming at — and does it put a human in the room (surgery 2.1, emergency medicine 2.5, primary care 2.9) or a pattern on a screen (pathology 4.7, radiology 4.9)? 2. Looking at the score table, which destination are you currently pointed at, and is it the one you'd choose on purpose? 3. Of the durable clinical strengths — physical examination and intuition, judgment under uncertainty, communication at the hardest moments, verification — which are you actively building, and which are you leaving to chance?

The move from here

4. What's the single most durable specialty you could realistically aim for from where you are now, and what would make you a credible candidate for it? 5. If you're still training: which rotations or electives put you closest to the physical, high-trust, decision-owning work — and

are you choosing them on purpose? 6. If you're in the Match or job-hunting: are you weighing the AI dimension in the specialty decision, or applying to whatever's competitive? Name the specialties you're actually chasing.

The AI question, practically

7. Of the five capabilities — verify, escalate, explain, recognize the limits, contribute — which one could you visibly demonstrate within the next few months? 8. Can you point to a moment where you caught a generated note, read or recommendation that was subtly wrong? If not, where's the easiest place to start building that skill?

Testing it against reality

9. **The most valuable single action:** find someone three to five years into practice — ideally in a specialty you're weighing — and ask what they actually did in year one, how the job has changed in the last three years, and whether that work still exists in the same form. 10. Moral injury — being unable to give the care you believe a patient needs because of time or staffing. Have you seen it on placement, and how do you think you'd cope with it over a career? It's what makes people leave a profession they still believe in.

This is analysis and informed judgment about a fast-moving situation, not a guarantee or a prediction. Where the evidence is contested we've said so, where our own score is challenged by the data we've flagged it, and where we might be wrong we've linked to the people making the opposite case. It's a starting point for a decision — not the decision itself.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology, not archived from published editions. Each year is scored on what was known at the time. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Medicine

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — what share of the working week could a capable current system already do to an acceptable standard?

RATING	ANCHOR
--------	--------

- | | |
|------|---|
| 0–2 | Almost nothing. The work is physical, relational or wholly novel. |
| 3–4 | Administrative surround only — notes, scheduling, correspondence. |
| 5–6 | A meaningful minority of the week, mostly documentation and lookup. |
| 7–8 | A large share, including tasks central to the junior version of the role. |
| 9–10 | Most of the described duties, at or near professional standard. |

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
--------	--------

- | | |
|------|---|
| 0–2 | Protected by statute or physical necessity. Training hours cannot be automated. |
| 3–4 | Stable. Some junior tasks absorbed; the apprenticeship survives. |
| 5–6 | Visible narrowing. Employers prefer experience; junior intake reduced. |
| 7–8 | Substantial. The work juniors learned on has largely gone. |
| 9–10 | The training layer and the automatable layer were the same layer. |

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
--------	--------

- | | |
|------|--|
| 0–2 | Fully screen-based. Location-independent. |
| 3–4 | Occasional presence; the core work is not physical. |
| 5–6 | Regular physical presence, in largely controlled conditions. |
| 7–8 | Substantially physical, with some variability of setting. |
| 9–10 | Irreducibly physical, in conditions that differ every time. |

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

SURGERY — 2.1 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.9	4.1	4.5	1.57
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	2.1 → 2.1

EMERGENCY MEDICINE — 2.5 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.3	4.8	5.5	1.92
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	9.0	9.0	9.0	0.15
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.2	9.2	9.2	0.08
				Total	2.5 → 2.5

PRIMARY CARE / FAMILY MEDICINE — 2.9 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.8	5.3	6.0	2.1
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	2.9 → 2.9

ANESTHESIOLOGY — 3.1 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.6	6.1	6.6	2.31
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	3.11 → 3.1

PSYCHIATRY — 3.7 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.0	6.7	7.4	2.59
Entry-path erosion	15%	1.2	1.5	1.8	0.27
Physical / embodied	15%	6.0	6.0	6.0	0.6
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	3.69 → 3.7

PATHOLOGY — 4.7 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.2	7.5	8.6	3.01
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	5.0	5.0	5.0	0.75
Human trust / accountability	15%	8.0	8.0	8.0	0.3
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	4.71 → 4.7

RADIOLOGY — 4.9 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.3	7.6	8.9	3.11
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	4.5	4.5	4.5	0.82
Human trust / accountability	15%	8.0	8.0	8.0	0.3
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	4.89 → 4.9

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Surgery

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	3.9	1.0	0.3	1.8
2025	4.1	1.2	0.3	1.9
Fall 2026	4.5	1.5	0.3	2.1

Movement decomposition: automatability contributed **+0.21** and entry-path erosion **+0.07**, for a total of **+0.3** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.